



电磁场与电磁波
Electromagnetic Field and Wave

其他介质分界面上 均匀平面波的反反射

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研讨：反/折射定律的再分析

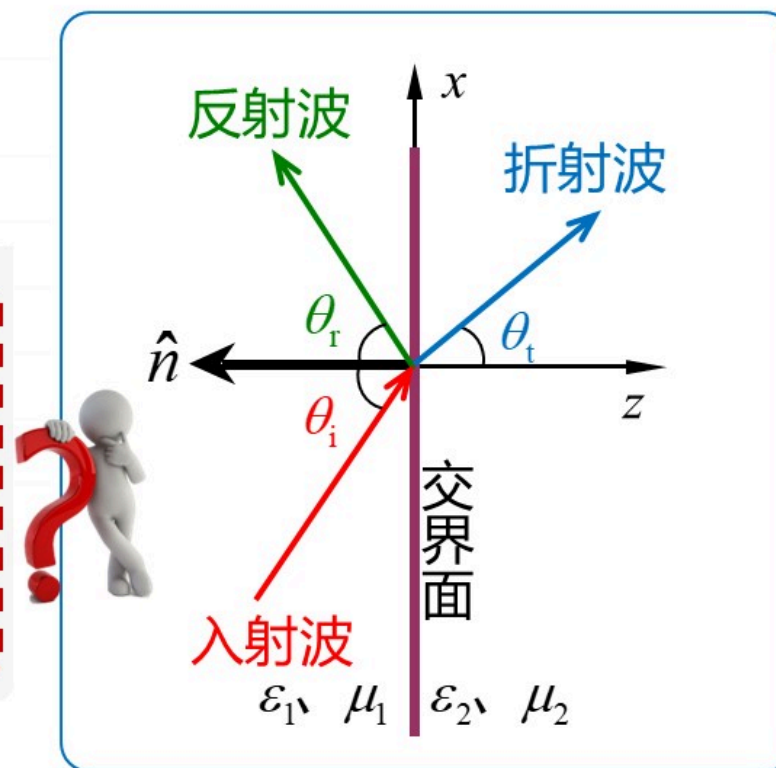


课前回顾

分析反折射问题的一般思路？

$$\begin{aligned}
 \vec{H}_i(\vec{r}) &= \hat{y} \frac{E_{i0}}{\eta_1} e^{-j\vec{k}_i \cdot \vec{r}} & \vec{k}_i &= k_1(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i) \\
 \vec{H}_r(\vec{r}) &= \hat{y} \frac{E_{r0}}{\eta_1} e^{-j\vec{k}_r \cdot \vec{r}} & \vec{k}_r &= k_1(\hat{x}\sin\theta_r - \hat{z}\cos\theta_r) \\
 \vec{H}_t(\vec{r}) &= \hat{y} \frac{E_{t0}}{\eta_2} e^{-j\vec{k}_t \cdot \vec{r}} & \vec{k}_t &= k_2(\hat{x}\sin\theta_t + \hat{z}\cos\theta_t)
 \end{aligned}$$

先假设场，再待定系数



研讨：反/折射定律的再分析

$$\vec{E}_i(\vec{r}, t) = \text{Re} \left[\vec{E}_{im} e^{j(\omega_i t - \vec{k}_i \cdot \vec{r})} \right]$$

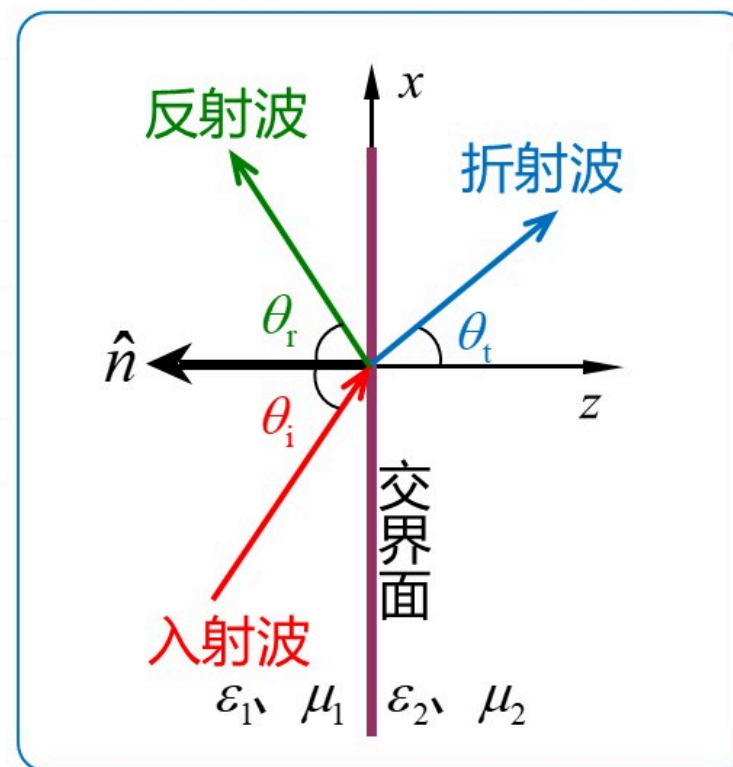
$$\vec{E}_r(\vec{r}, t) = \text{Re} \left[\vec{E}_{rm} e^{j(\omega_r t - \vec{k}_r \cdot \vec{r})} \right]$$

$$\vec{E}_t(\vec{r}, t) = \text{Re} \left[\vec{E}_{tm} e^{j(\omega_t t - \vec{k}_t \cdot \vec{r})} \right]$$

$$k_i = \omega_i \sqrt{\epsilon_1 \mu_1}$$

$$k_r = \omega_r \sqrt{\epsilon_1 \mu_1}$$

$$k_t = \omega_t \sqrt{\epsilon_2 \mu_2}$$

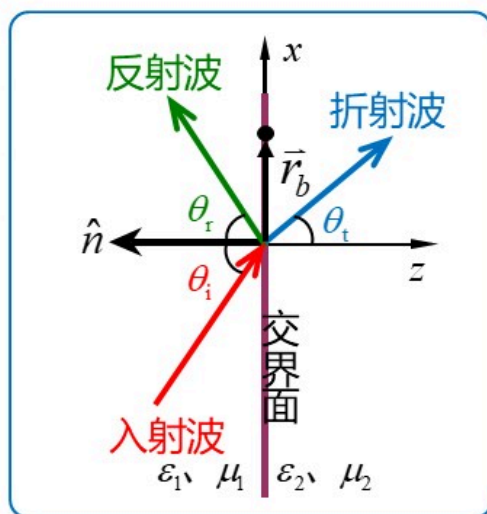


研讨：反/折射定律的再分析

$$\vec{E}_i(\vec{r}, t) = \text{Re} \left[\vec{E}_{im} e^{j(\omega t - \vec{k}_i \cdot \vec{r})} \right]$$

$$\vec{E}_r(\vec{r}, t) = \text{Re} \left[\vec{E}_{rm} e^{j(\omega t - \vec{k}_r \cdot \vec{r})} \right]$$

$$\vec{E}_t(\vec{r}, t) = \text{Re} \left[\vec{E}_{tm} e^{j(\omega t - \vec{k}_t \cdot \vec{r})} \right]$$



边界条件：

$$\hat{n} \times (\vec{E}_1 - \vec{E}_2) = 0 \quad \text{或} \quad E_{1t} - E_{2t} = 0$$

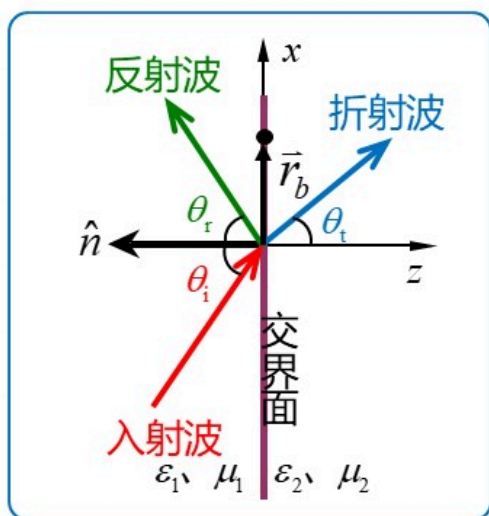
$$\Rightarrow \vec{E}_{it} \Big|_{z=0} + \vec{E}_{rt} \Big|_{z=0} = \vec{E}_{tt} \Big|_{z=0}$$

$$\text{Re} \left[\vec{E}_{im} e^{j(\omega t - \vec{k}_i \cdot \vec{r}_b)} + \vec{E}_{rm} e^{j(\omega t - \vec{k}_r \cdot \vec{r}_b)} \right]$$

$$= \text{Re} \left[\vec{E}_{tm} e^{j(\omega t - \vec{k}_t \cdot \vec{r}_b)} \right]$$

$$\Rightarrow \begin{cases} \omega_i = \omega_r = \omega_t = \omega \\ \vec{k}_r \cdot \vec{r}_b = \vec{k}_t \cdot \vec{r}_b = \vec{k}_i \cdot \vec{r}_b \end{cases}$$

研讨：反/折射定律的再分析



$$\vec{k}_r \cdot \vec{r}_b = \vec{k}_t \cdot \vec{r}_b = \vec{k}_i \cdot \vec{r}_b$$

$$\hat{n} \times (\hat{n} \times \vec{r}_b) = \hat{n}(\hat{n} \cdot \vec{r}_b) - (\hat{n} \cdot \hat{n}) \vec{r}_b$$

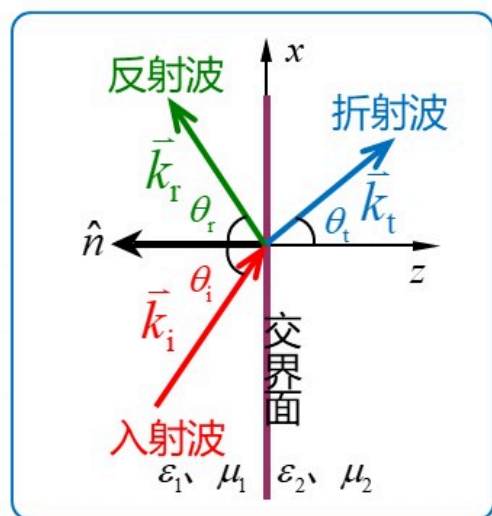
$$\Rightarrow \vec{r}_b = -\hat{n} \times (\hat{n} \times \vec{r}_b)$$

$$\Rightarrow \vec{k}_i \cdot \vec{r}_b = -\vec{k}_i \cdot [\hat{n} \times (\hat{n} \times \vec{r}_b)] = -(\hat{n} \times \vec{r}_b) \cdot (\vec{k}_i \times \hat{n})$$

$$(\hat{n} \times \vec{r}_b) \cdot (\vec{k}_r \times \hat{n}) = (\hat{n} \times \vec{r}_b) \cdot (\vec{k}_t \times \hat{n}) = (\hat{n} \times \vec{r}_b) \cdot (\vec{k}_i \times \hat{n})$$

$$\Rightarrow \vec{k}_r \times \hat{n} = \vec{k}_t \times \hat{n} = \vec{k}_i \times \hat{n}$$

研讨：反/折射定律的再分析



结论1: 传播矢量 $\vec{k}_i$ 、 $\vec{k}_r$ 、 $\vec{k}_t$ 共面

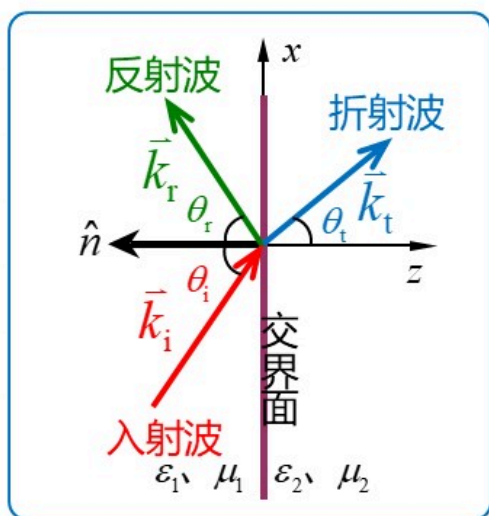
$$\vec{k}_r \times \hat{n} = \vec{k}_t \times \hat{n} = \vec{k}_i \times \hat{n}$$

$$\vec{k}_i = k_1 (\hat{x} \sin \theta_i + \hat{z} \cos \theta_i)$$

$$\vec{k}_r = k_1 (\hat{x} \sin \theta_r - \hat{z} \cos \theta_r)$$

$$\vec{k}_t = k_2 (\hat{x} \sin \theta_t + \hat{z} \cos \theta_t)$$

研讨：反/折射定律的再分析



结论2：反射角等于入射角（反射定律）

$$\underline{\vec{k}_r \times \hat{n}} = \underline{\vec{k}_t \times \hat{n}} = \underline{\vec{k}_i \times \hat{n}}$$

$$\vec{k}_i = k_1(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i)$$

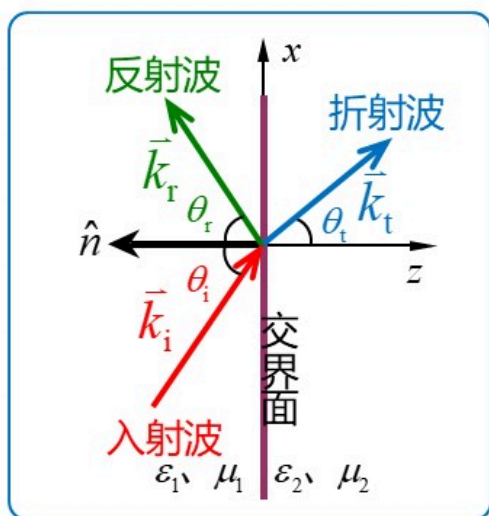
$$\vec{k}_r = k_1(\hat{x}\sin\theta_r - \hat{z}\cos\theta_r)$$

$$\Rightarrow k_1\sin\theta_i = k_1\sin\theta_r$$

$$\Rightarrow \theta_r = \theta_i$$

研讨：反/折射定律的再分析

结论3：折射定律（Snell定律）



$$\vec{k}_r \times \hat{n} = \vec{k}_t \times \hat{n} = \vec{k}_i \times \hat{n}$$

$$\vec{k}_i = k_1 (\hat{x} \sin \theta_i + \hat{z} \cos \theta_i)$$

$$\vec{k}_t = k_2 (\hat{x} \sin \theta_t + \hat{z} \cos \theta_t)$$

$$\Rightarrow k_1 \sin \theta_i = k_2 \sin \theta_t$$

主要内容

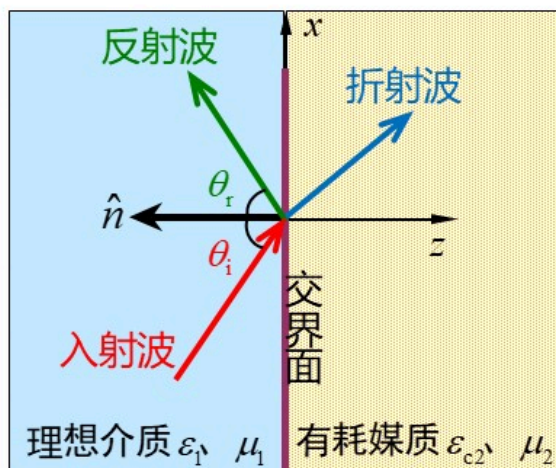
一

电磁波从理想介质向有耗媒质入射

二

电磁波从理想介质向理想导体入射

一、电磁波从理想介质向有耗媒质入射



研究方法: **形式代换**

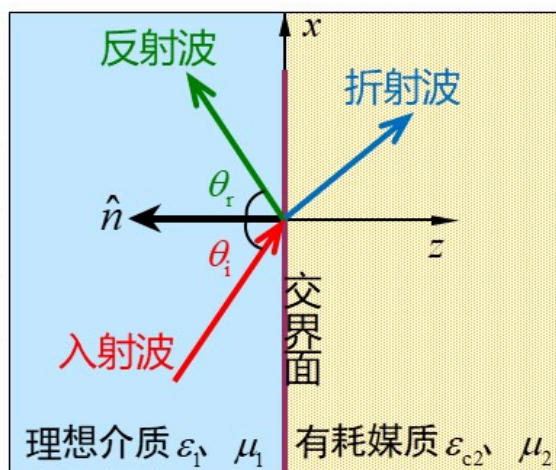
问题一: 反射定律、折射定律如何处理?

问题二: 菲涅耳公式如何处理?



一、电磁波从理想介质向有耗媒质入射

问题一：反射定律、折射定律如何处理？



$$\underline{\vec{k}_r \times \hat{n}} = \underline{\vec{k}_t \times \hat{n}} = \underline{\vec{k}_i \times \hat{n}}$$

1. 反射定律: $\theta_r = \theta_i$

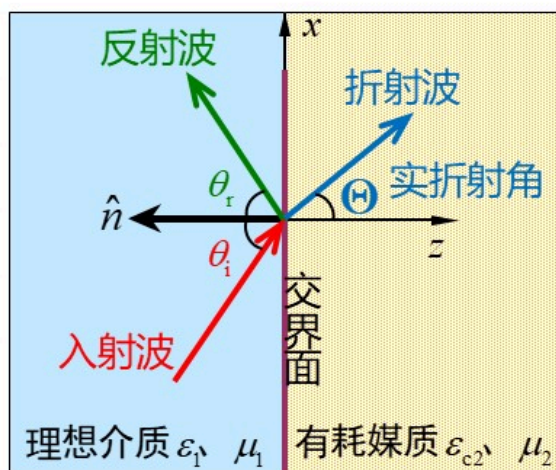
2. 折射定律?

$$k_1 \sin \theta_i = k_2 \sin \tilde{\theta}_t$$



一、电磁波从理想介质向有耗媒质入射

2. 折射定律



$$\hat{n} = -\hat{z}$$

引入**复折射角** $\tilde{\theta}_t$ ，则透射波传播矢量可表示为

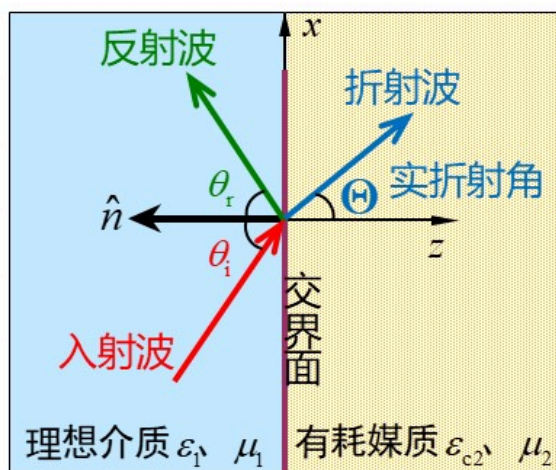
$$\vec{k}_t = k_2 (\hat{x} \sin \tilde{\theta}_t + \hat{z} \cos \tilde{\theta}_t) = \vec{\beta} - j \vec{\alpha}$$

分界面上：

$$\because \vec{k}_t \times \vec{n} = \vec{k}_i \times \vec{n}$$

$$\therefore \begin{cases} \vec{k}_i \times \hat{z} = \vec{\beta} \times \hat{z} \\ \vec{\alpha} \times \hat{z} = 0 \end{cases}$$

一、电磁波从理想介质向有耗媒质入射



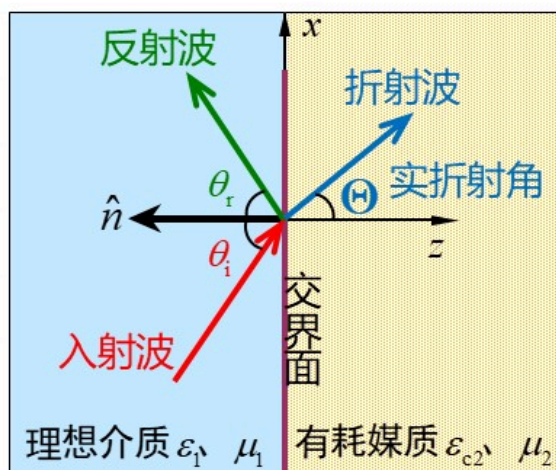
$$\begin{cases} \vec{k}_i \times \hat{z} = \vec{\beta} \times \hat{z} \\ \vec{\alpha} \times \hat{z} = 0 \end{cases}$$

设矢量 $\vec{\beta}$ 与交界面法向的夹角为 Θ ，可得

$$\begin{cases} k_1 \sin \theta_i = \beta \sin \Theta \\ \vec{\alpha} = \alpha \hat{z} \end{cases}$$

通常称 Θ 为透射波的**实折射角**

一、电磁波从理想介质向有耗媒质入射



由于 $\vec{\beta}$ 与 $\vec{k}_i$ 共面

$$\text{则 } \vec{\beta} = \hat{x}\beta\sin\Theta + \hat{z}\beta\cos\Theta$$

$$\begin{aligned}\vec{k}_t &= \vec{\beta} - j\vec{\alpha} = \hat{x}\beta\sin\Theta + \hat{z}(\beta\cos\Theta - j\alpha) \\ &= \hat{x}k_1\sin\theta_i + \hat{z}(\beta\cos\Theta - j\alpha)\end{aligned}$$

$$\Rightarrow k_1\sin\theta_i = \beta\sin\Theta = k_2\sin\tilde{\theta}_t \quad \text{复折射定律}$$

结论

媒质2中的折射波是一个沿 $\hat{z}$ 方向衰减、沿 $\vec{\beta}$ 方向传播的非均匀平面波

一、电磁波从理想介质向有耗媒质入射

问题二：菲涅耳公式如何处理？

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$\Gamma_{//} = \frac{\eta_1 \cos \theta_i - \eta_2 \cos \theta_t}{\eta_1 \cos \theta_i + \eta_2 \cos \theta_t}$$

$$\tau_{//} = \frac{2\eta_2 \cos \theta_i}{\eta_1 \cos \theta_i + \eta_2 \cos \theta_t}$$

$\Rightarrow$

$\Rightarrow$

$$\Gamma_{\perp} = \frac{\eta_{c2} \cos \theta_i - \eta_1 \cos \tilde{\theta}_t}{\eta_{c2} \cos \theta_i + \eta_1 \cos \tilde{\theta}_t}$$

$$\tau_{\perp} = \frac{2\eta_{c2} \cos \theta_i}{\eta_{c2} \cos \theta_i + \eta_1 \cos \tilde{\theta}_t}$$

$$\Gamma_{//} = \frac{\eta_1 \cos \theta_i - \eta_{c2} \cos \tilde{\theta}_t}{\eta_1 \cos \theta_i + \eta_{c2} \cos \tilde{\theta}_t}$$

$$\tau_{//} = \frac{2\eta_{c2} \cos \theta_i}{\eta_1 \cos \theta_i + \eta_{c2} \cos \tilde{\theta}_t}$$

形式代换、附加相移

二、电磁波从理想介质向理想导体入射

向理想导体 ($\sigma = \infty$) 入射时的菲涅耳公式

$$\eta_{c2} = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon - j\frac{\sigma}{\omega}}} \rightarrow 0$$

垂直极化入射

$$\Gamma_{\perp} = \frac{\eta_{c2} \cos \theta_i - \eta_1 \cos \tilde{\theta}_t}{\eta_{c2} \cos \theta_i + \eta_1 \cos \tilde{\theta}_t} = -1$$

$$\tau_{\perp} = \frac{2\eta_{c2} \cos \theta_i}{\eta_{c2} \cos \theta_i + \eta_1 \cos \tilde{\theta}_t} = 0$$

平行极化入射:

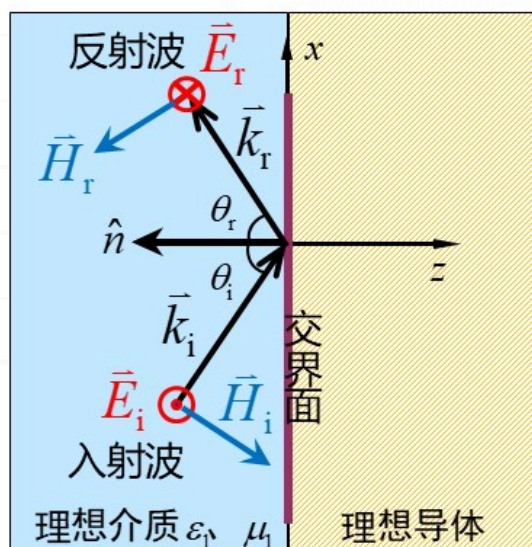
$$\Gamma_{//} = \frac{\eta_1 \cos \theta_i - \eta_{c2} \cos \tilde{\theta}_t}{\eta_1 \cos \theta_i + \eta_{c2} \cos \tilde{\theta}_t} = 1$$

$$\tau_{//} = \frac{2\eta_{c2} \cos \theta_i}{\eta_1 \cos \theta_i + \eta_{c2} \cos \tilde{\theta}_t} = 0$$

结论: 电磁波被全反射

二、电磁波从理想介质向理想导体入射

1 垂直极化入射



$$\begin{aligned}\vec{E}_i(\vec{r}) &= \hat{y}E_{i0}e^{-jk_1\vec{r}} \\ &= \hat{y}E_{i0}e^{-jk_1(x\sin\theta_i+z\cos\theta_i)}\end{aligned}$$

$$\begin{aligned}\vec{E}_r(\vec{r}) &= \hat{y}\Gamma_{\perp}E_{i0}e^{-jk_1\vec{r}} \\ &= -\hat{y}E_{i0}e^{-jk_1(x\sin\theta_i-z\cos\theta_i)}\end{aligned}$$

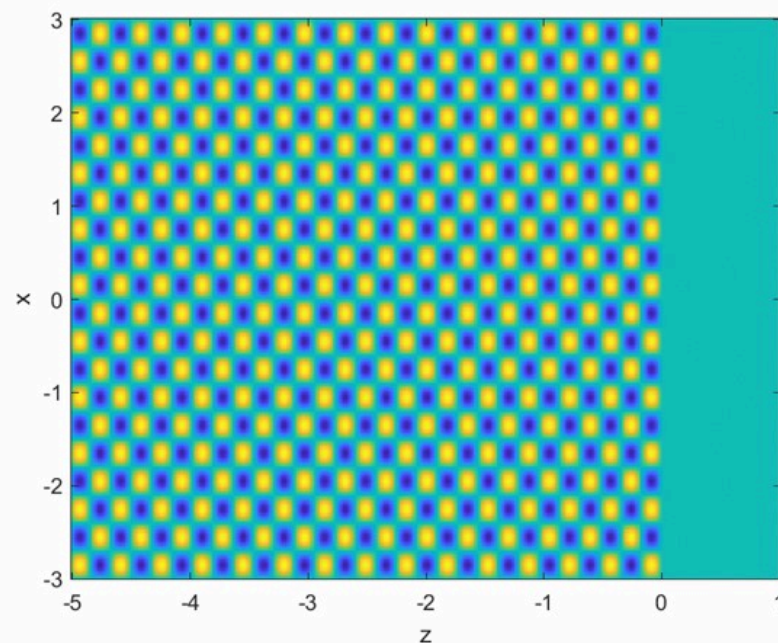
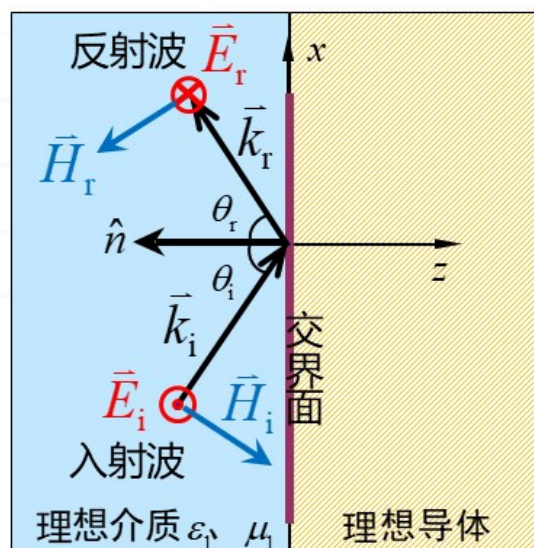
$$\vec{H}_i(\vec{r}) = (-\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)\frac{E_{i0}}{\eta_1}e^{-jk_1(x\sin\theta_i+z\cos\theta_i)}$$

$$\vec{H}_r(\vec{r}) = (-\hat{x}\cos\theta_i - \hat{z}\sin\theta_i)\frac{E_{i0}}{\eta_1}e^{-jk_1(x\sin\theta_i-z\cos\theta_i)}$$

二、电磁波从理想介质向理想导体入射

媒质1中的合成场

$$\vec{E}_1(\vec{r}) = \vec{E}_i(\vec{r}) + \vec{E}_r(\vec{r}) = -\hat{y}2jE_{i0}\sin(k_1z\cos\theta_i)e^{-jk_1x\sin\theta_i}$$



二、电磁波从理想介质向理想导体入射

媒质1中的合成场

$$\vec{E}_1(\vec{r}) = \vec{E}_i(\vec{r}) + \vec{E}_r(\vec{r}) = -\hat{y}2jE_{i0}\sin(k_1z\cos\theta_i)e^{-jk_1x\sin\theta_i}$$

$$\vec{H}_1(\vec{r}) = \vec{H}_i(\vec{r}) + \vec{H}_r(\vec{r})$$

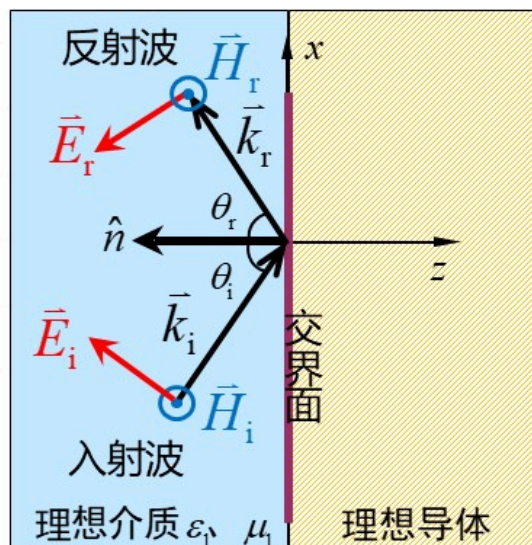
$$= \left[-\hat{x}\cos\theta_i\cos(k_1z\cos\theta_i) - \hat{z}j\sin\theta_i\sin(k_1z\cos\theta_i) \right] \frac{2E_{i0}}{\eta_1} e^{-jk_1x\sin\theta_i}$$

结论

- (1) 合成波沿平行于分界面的方向 (x 方向) 传播
- (2) 合成波在垂直于导体表面的方向 (z 方向) 呈驻波分布
- (3) 合成波是非均匀平面波
- (4) 合成波是横电波 (TE波)

二、电磁波从理想介质向理想导体入射

2 平行极化入射



$$\vec{H}_i(\vec{r}) = \hat{y} \frac{E_{i0}}{\eta_1} e^{-j\vec{k}_i \cdot \vec{r}}$$

$$= \hat{y} \frac{E_{i0}}{\eta_1} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}$$

$$\vec{H}_r(\vec{r}) = \hat{y} \Gamma_{//} \frac{E_{i0}}{\eta_1} e^{-j\vec{k}_r \cdot \vec{r}}$$

$$= \hat{y} \frac{E_{i0}}{\eta_1} e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}$$

$$\vec{E}_i(\vec{r}) = (\hat{x}\cos\theta_i - \hat{z}\sin\theta_i) E_{i0} e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}$$

$$\vec{E}_r(\vec{r}) = (-\hat{x}\cos\theta_i - \hat{z}\sin\theta_i) E_{i0} e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}$$

二、电磁波从理想介质向理想导体入射

媒质1中的合成场

$$\vec{H}_1 = \vec{H}_i(\vec{r}) + \vec{H}_r(\vec{r}) = \hat{y} \frac{2E_{i0}}{\eta_1} \cos(k_1 z \cos \theta_i) e^{-jk_1 x \sin \theta_i}$$

$$\vec{E}_1(\vec{r}) = \vec{E}_i(\vec{r}) + \vec{E}_r(\vec{r})$$

$$= \left[-\hat{x} j \cos \theta_i \sin(k_1 z \cos \theta_i) - \hat{z} \sin \theta_i \cos(k_1 z \cos \theta_i) \right] 2E_{i0} e^{-jk_1 x \sin \theta_i}$$

结论

- (1) 合成波沿平行于分界面的方向 (x 方向) 传播
- (2) 合成波在垂直于导体表面的方向 (z 方向) 呈驻波分布
- (3) 合成波是非均匀平面波
- (4) 合成波是横磁波 (TM波)

课堂总结



- (1) 电磁波在理想介质与有耗媒质分界面上的反与折射
形式代换、附加相移
- (2) 电磁波向理想导体入射时的全反射现象

$$\Gamma_{\perp} = \frac{\eta_{c2} \cos \theta_1 - \eta_1 \cos \tilde{\theta}_t}{\eta_{c2} \cos \theta_1 + \eta_1 \cos \tilde{\theta}_t} = -1$$
$$\tau_{\perp} = \frac{2\eta_{c2} \cos \theta_1}{\eta_{c2} \cos \theta_1 + \eta_1 \cos \tilde{\theta}_t} = 0$$

$$\Gamma_{//} = \frac{\eta_1 \cos \theta_1 - \eta_{c2} \cos \tilde{\theta}_t}{\eta_1 \cos \theta_1 + \eta_{c2} \cos \tilde{\theta}_t} = 1$$
$$\tau_{//} = \frac{2\eta_{c2} \cos \theta_1}{\eta_1 \cos \theta_1 + \eta_{c2} \cos \tilde{\theta}_t} = 0$$